



Lab 06: Departmental Subnetting

IT · FINANCE · HR PCs · Switch3 (2960) · Router0 (2911)

Lab #06

Topic: Subnetting &
Department Networks

Tool: Cisco Packet Tracer

Cert: CompTIA Network+

1. Lab Objectives

- Create three departmental subnets for IT, Finance and HR on one physical switch.
- Connect all three department PCs to Switch3 (2960-24TT) on Fa0/1, Fa0/2, Fa0/3.
- Connect Switch3 to Router0 (2911) on Fa0/4 → Gig0/0.
- Configure router subinterfaces (Router-on-a-Stick) for inter-department routing.
- Verify inter-department communication using ping.

2. Network Topology

Switch3 (2960-24TT) is the central switch. IT PC connects on Fa0/1, Finance PC on Fa0/2, HR PC on Fa0/3. Switch3 uplinks to Router0 (2911) via Fa0/4 → Gig0/0. The router uses sub-interfaces for each department VLAN.

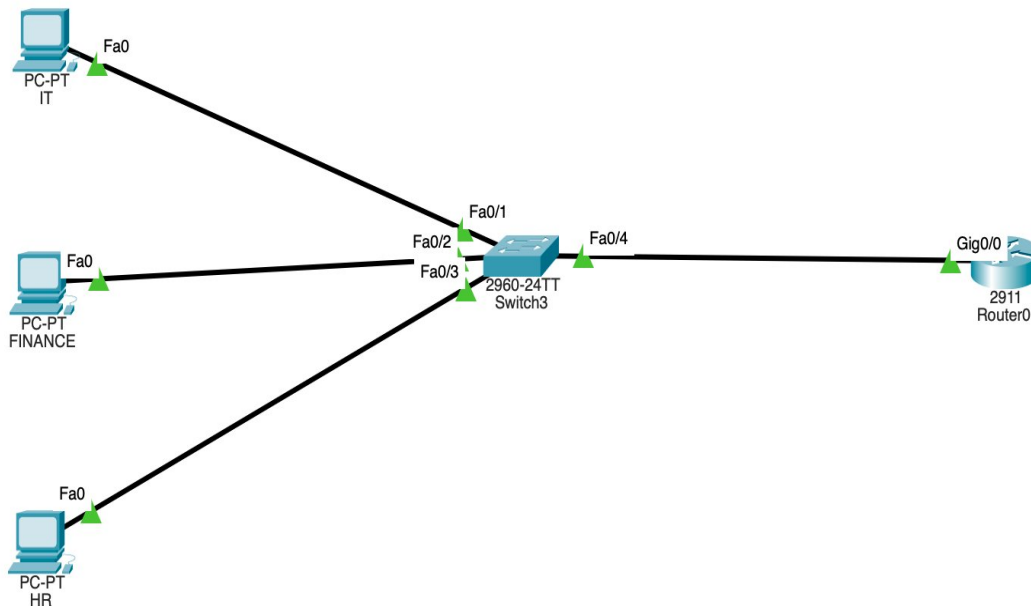


Figure 1 – IT, Finance, HR PCs on Switch3 connected to Router0 for inter-dept routing

Department	PC Name	Switch Port	VLAN	IP Address	Gateway
IT	IT	Fa0/1	VLAN 10	192.168.10.10	192.168.10.1



Finance	FINANCE	Fa0/2	VLAN 20	192.168.20.10	192.168.20.1
HR	HR	Fa0/3	VLAN 30	192.168.30.10	192.168.30.1
Router0	—	Gig0/0 (trunk)	All	Sub-interfaces	—

3. Step-by-Step Configuration

3.1 Switch3 – Create VLANs and Assign Ports

```
Switch3> enable
Switch3# configure terminal
Switch3(config)# vlan 10
Switch3(config-vlan)# name IT
Switch3(config-vlan)# exit
Switch3(config)# vlan 20
Switch3(config-vlan)# name FINANCE
Switch3(config-vlan)# exit
Switch3(config)# vlan 30
Switch3(config-vlan)# name HR
Switch3(config-vlan)# exit
Switch3(config)# interface Fa0/1
Switch3(config-if)# switchport mode access
Switch3(config-if)# switchport access vlan 10
Switch3(config-if)# exit
Switch3(config)# interface Fa0/2
Switch3(config-if)# switchport mode access
Switch3(config-if)# switchport access vlan 20
Switch3(config-if)# exit
Switch3(config)# interface Fa0/3
Switch3(config-if)# switchport mode access
Switch3(config-if)# switchport access vlan 30
Switch3(config-if)# exit
Switch3(config)# interface Fa0/4
Switch3(config-if)# switchport mode trunk
Switch3(config-if)# exit
Switch3# write memory
```

3.2 Router0 – Sub-interfaces (Router-on-a-Stick)

```
Router0(config)# interface GigabitEthernet0/0
Router0(config-if)# no shutdown
Router0(config-if)# exit
Router0(config)# interface GigabitEthernet0/0.10
```



```
Router0(config-subif)# encapsulation dot1Q 10
Router0(config-subif)# ip address 192.168.10.1 255.255.255.0
Router0(config-subif)# exit
Router0(config)# interface GigabitEthernet0/0.20
Router0(config-subif)# encapsulation dot1Q 20
Router0(config-subif)# ip address 192.168.20.1 255.255.255.0
Router0(config-subif)# exit
Router0(config)# interface GigabitEthernet0/0.30
Router0(config-subif)# encapsulation dot1Q 30
Router0(config-subif)# ip address 192.168.30.1 255.255.255.0
Router0(config-subif)# exit
Router0# write memory
```

4. Verification

```
IT> ping 192.168.20.10      ! IT to Finance
IT> ping 192.168.30.10      ! IT to HR
Switch3# show vlan brief    ! Verify VLAN assignments
Switch3# show interfaces trunk ! Fa0/4 should be trunk
```

5. Key Takeaways

- Departmental subnetting isolates broadcast domains and improves network security.
- Router-on-a-Stick uses a single trunk link with 802.1Q sub-interfaces — efficient and cost-effective.
- Each department PC uses its sub-interface IP as the default gateway.
- The trunk port on the switch must be configured before the router sub-interfaces will work.
- Real-world networks often use a Layer 3 switch instead of Router-on-a-Stick for better performance.